

E-DENTIST:

Comprehensive SRS & Analysis Document

Course: CSE 6159 (M) SSA

Submitted to:

Dr. Suman Ahmmed

Head and Associate Professor, Dept. of CSE, UIU

Prepared By:

Md. Ahsan Kabir (0122520016)

Abu Sadat Al Rafi (0122230017)

Abu Ekhtiar Tawhid (0122230018)



United International University

Dhaka, Bangladesh

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1. Introduction

E-Dentist is a specialized tele-dentistry platform designed to deliver accessible, affordable, and high-quality oral healthcare services across Bangladesh through the effective use of modern digital technologies. By integrating web-based and mobile applications, the platform enables patients to receive remote dental consultations, intelligent risk-based triage, digital prescriptions, automated reminders, and clinic navigation services. The system aims to bridge the persistent gap between patients and qualified dental professionals by overcoming geographical, economic, and infrastructural barriers that limit access to timely oral healthcare.

Oral healthcare remains a significantly underserved component of the healthcare system in Bangladesh, particularly in rural and semi-urban regions. Factors such as limited availability of dental professionals, low public awareness of preventive dental care, high out-of-pocket treatment costs, and inadequate healthcare infrastructure contribute to delayed diagnosis and poor treatment outcomes. The situation is further exacerbated during public health emergencies such as the COVID-19 pandemic, where physical clinic visits become risky or impractical. E-Dentist addresses these challenges by enabling safe, remote interaction between patients and dentists, reducing unnecessary travel, and supporting continuity of care during crisis situations.

Oral health is closely linked to overall physical well-being and plays a critical role in the prevention and management of systemic conditions such as cardiovascular diseases, diabetes, and respiratory infections. In Bangladesh, it is estimated that nearly 80% of the population suffers from at least one form of oral disease, while the dentist-to-patient ratio remains alarmingly low at approximately one dentist per 200,000 individuals. These disparities are most pronounced in rural areas, where access to specialized dental services is minimal. Although several telemedicine platforms currently operate in Bangladesh, they primarily focus on generalized medical consultations and lack dentistry-specific capabilities, such as visual oral diagnostics, interactive dental models, oral risk scoring, and structured dental triage workflows. To address these limitations, E-Dentist proposes a comprehensive and dental-focused digital ecosystem that combines both synchronous (real-time video consultations) and asynchronous (store-and-forward exchange of clinical data such as images and records) modes of tele-dentistry. The platform leverages a cloud-based, scalable architecture to ensure interoperability, reliability, and nationwide accessibility. Key features such as intelligent symptom assessment, risk-based appointment prioritization, and offline-capable clinic navigation are designed to meet the unique healthcare realities of Bangladesh, particularly for underserved populations.

This document presents a complete Software Requirements Specification (SRS) for the E-Dentist system. It includes detailed functional and non-functional requirements, system architecture analysis, UML and behavioral diagrams, algorithmic specifications, and requirements traceability. Furthermore, the document is structured in full compliance with Capability Maturity Model Integration (CMMI) Development (DEV) Version 3.0 standards, aligned with CMMI Maturity Level 3 ("Defined"). All relevant Practice Areas (PAs) are systematically mapped, supported by acceptance criteria, verification mechanisms, and an expanded traceability matrix.

2. Motivation

Bangladesh faces significant challenges in oral healthcare delivery, including limited dental specialists in rural areas, lack of awareness, long wait times, minimal digitization, and heightened risks during pandemics like COVID-19. National statistics indicate that rural populations are less knowledgeable about oral hygiene, with district-level health complexes often lacking dental facilities. Telemedicine solutions exist (e.g., DocTime, Digital Hospital, Seba Ghar), but none provide a dedicated, fully integrated dental care platform handling remote triage, video consultation, and oral risk assessment.

E-Dentist is motivated by the need to:

- Reduce inequity in dental care access, particularly for underserved rural and semi-urban communities.
- Support remote diagnosis through video and asynchronous consultations, minimizing infection risks for both patients and dentists.
- Offer a structured, AI-assisted triage system to prioritize urgent cases and promote preventive care.
- Ensure continuous oral health monitoring, including self-checkups and community campaigns.
- Digitalize prescriptions and follow-up schedules to improve adherence and outcomes.

Literature highlights tele-dentistry's potential to extend expertise cost-effectively, with studies showing positive perceptions among specialists and patients. During COVID-19, tele-dentistry enabled continued practice while reducing exposure, underscoring its role in resilient healthcare systems.

3. Vision

E-Dentist aims to become Bangladesh's first comprehensive tele-dentistry ecosystem by enabling:

- Smart, risk-based scheduling ensuring urgent cases receive priority.
- Interactive 3D tooth models for intuitive patient-dentist communication.
- Integrated oral health monitoring for communities and schools.
- Offline-capable clinic navigation ensuring accessibility.
- Cloud-based scalability to serve millions.

The long-term vision includes interoperability with national health systems, integration with insurance providers, machine learning for advanced symptom analysis, and virtual health campaigns for disadvantaged groups. This aligns with global tele-dentistry benefits, such as improved access for people with disabilities and time efficiency for practitioners.

4. Literature Review

Tele-dentistry, defined as the application of telecommunication technologies to deliver dental care, consultation, education, and public awareness, has been extensively studied in both global and local healthcare contexts. Existing research highlights its potential to improve access to oral healthcare services, particularly in regions facing shortages of dental professionals and infrastructural limitations. However, these studies also reveal persistent gaps in the provision of specialized, dentistry-focused digital solutions.

In the Bangladeshi context, telemedicine platforms reviewed by Chowdhury et al. (2018) primarily focus on generalized healthcare services, offering features such as video consultations, mobile health (mHealth) support, and remote physician access. While these platforms improve healthcare accessibility, they lack essential dental-specific functionalities, including visual oral diagnostics, structured dental symptom assessment, and patient education tools tailored to oral health. As a result, dentistry remains underserved within existing telemedicine ecosystems, limiting the effectiveness of remote dental care delivery.

Global systematic reviews, such as the work by Estai et al. (2016), demonstrate that tele-dentistry significantly contributes to early diagnosis, cost reduction, and continuity of care, particularly for rural and underserved populations. These studies identify two primary operational modes: synchronous tele-dentistry, which enables real-time interaction between patients and dental professionals through video consultations, and asynchronous tele-dentistry, which supports store-and-forward transmission of clinical data such as images and records for diagnosis and treatment planning. Together, these modes form the foundation of effective remote dental care.

Several Bangladesh-focused studies address individual components relevant to tele-dentistry but remain fragmented in scope. Rahman and Karim (2016) explore mobile-based medication reminder systems aimed at improving patient adherence, while Deb et al. (2016) investigate offline navigation systems for accessing healthcare facilities in low-connectivity environments. Although these solutions address critical challenges, they operate as isolated systems rather than an integrated dental care platform.

Similarly, research on online appointment scheduling systems (Sherly et al., 2017) and Wi-Fi-based video calling platforms (Shrideep et al., 2014) provides technological building blocks for telehealth applications. However, these studies do not incorporate dental-specific innovations such as interactive 3D tooth models, oral risk scoring algorithms, or intelligent triage mechanisms that are essential for accurate dental assessment and prioritization.

Despite these limitations, perception-based studies report positive attitudes toward tele-dentistry. Dental professionals regard it as an innovative and safe approach, particularly during public health emergencies such as the COVID-19 pandemic (Deshpande et al., 2020). Patients also express satisfaction with improved accessibility, reduced travel requirements, and faster consultations (Rahman et al., 2020). These findings underscore both the feasibility and acceptance of tele-dentistry solutions.

This literature review directly informs the design of the E-Dentist system by identifying the need for an integrated, dental-specific telehealth platform. E-Dentist addresses these gaps by combining synchronous and asynchronous consultation models, intelligent risk-based triage, digital prescriptions, automated reminders, and offline clinic navigation. By unifying these features within a single ecosystem, the proposed system aims to deliver equitable, efficient, and scalable oral healthcare services aligned with the healthcare realities of Bangladesh.

5. Detailed Information Gathering

Inputs for the E-Dentist system were systematically gathered through an extensive review of academic literature, empirical research studies, and an analysis of existing telemedicine and mHealth platforms currently used in Bangladesh and comparable developing regions. Peer-reviewed journals, conference papers, and case studies were examined to identify best practices, technological limitations, and emerging trends in tele-dentistry and digital healthcare delivery.

In addition to literature analysis, functional and non-functional insights were derived from studying existing telemedicine systems to understand real-world workflows, user interaction patterns, and service gaps—particularly in relation to dental care. This comparative analysis helped identify missing dental-specific features such as structured oral triage, risk assessment mechanisms, and visual diagnostic support.

The methodology employed both adaptive and quantitative approaches. The adaptive approach allowed system requirements to evolve based on observed limitations, stakeholder needs, and contextual constraints such as connectivity and digital literacy. Quantitative considerations, including response time expectations, user load capacity, and scheduling efficiency, informed performance, scalability, and reliability requirements.

Furthermore, the information gathering process focused on both synchronous (real-time) and asynchronous (store-and-forward) tele-dentistry models. Real-time interactions support immediate consultation and emergency triage, while asynchronous data exchange enables symptom reporting, image sharing, and treatment planning in low-bandwidth or intermittent connectivity environments. This dual-model focus ensures that the proposed system remains flexible, inclusive, and effective across diverse usage scenarios.

Overall, this structured information-gathering methodology ensures that the E-Dentist system requirements are evidence-based, context-aware, and aligned with established software engineering and healthcare informatics standards.

5.1 Stakeholder Identification

- Patients – require consultations, prescriptions, reminders.
- Dentists – need tools for triage, video calls, prescription, schedule management.
- Administrators – maintain system, users, reporting.
- Pharmacies – receive e-prescriptions (future scope).

5.2 Existing System Analysis

General telemedicine (e.g., DocTime, Digital Hospital, Seba Ghar) supports video calls but lacks:

- Dental-specific triage,
- Oral risk scoring,
- 3D tooth visualization,
- Dentist-specialization mapping,
- Dental emergency routing.

Research-based systems have explored components like advanced scheduling, OCR-based reminders, offline navigation, and Wi-Fi video calling, but no integrated dental solution exists.

Comparison Table:

Feature	General Telemedicine (DocTime, Digital Hospital, Seba Ghar)	Research-Based Systems	E-Dentist	Gap Addressed
Video Consultation	Yes (general medical focus)	Yes (Wi-Fi / prototype level)	Yes (dental-specific)	Dental-oriented consultation workflow
Dental-Specific Triage	No	Partial (rule-based / experimental)	Yes	Proper prioritization of dental cases
Oral Risk Scoring	No	Partial	Yes	Objective urgency assessment
3D Tooth Visualization	No	No	Yes	Accurate symptom localization
Dentist–Specialization Mapping	No	Partial	Yes	Correct specialist assignment
Dental Emergency Routing	No	No	Yes	Immediate response to urgent cases
Advanced Scheduling Logic	Basic (first-come)	Yes	Yes	Priority-based scheduling
Medication / OCR Reminders	Limited	Yes	Yes	Improved treatment adherence
Offline Clinic Navigation	Limited	Yes	Yes	Accessibility in rural areas
End-to-End Dental Integration	No	No	Yes	Unified tele-dentistry ecosystem

5.3 Functional Needs Identified

- Online registration & identity verification.
- Symptom entry with guided questionnaire.
- Algorithm-based risk calculation.
- Online scheduling with priority logic.
- Real-time and asynchronous consultation.

- Digital prescription generation.
- Medication reminders.
- Clinic finder with offline capability.
- Emergency dentist on call.

5.4 Constraints

- Variable internet quality across Bangladesh.
- Limited digital literacy among rural users.
- Need for secure storage compliant with health data regulations.
- Pandemic resilience.

6. Analysis, Feature List, and Site Map

6.1 Feature List

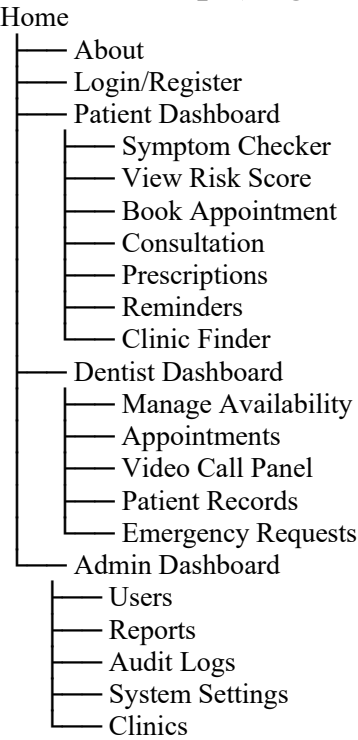
Core Modules:

- User & Access Management
- Symptom Checker & Risk Assessment
- Oral Health Monitoring
- Appointment & Scheduling
- Video/Asynchronous Consultation
- Prescription Management
- Reminder Engine
- Clinic Finder (Offline Support)
- Emergency Support
- Admin Dashboard

Novel Features from Analysis:

- Dental Specialization: Workflows tailored for oral diseases.
- Interactive 3D Dental Model: Visual symptom identification.
- Intelligent Oral Risk Analyzer: Algorithm for triage.
- Seamless Service Integration: Full care pathway.
- Offline-Capable Navigation: For rural access.
- Cloud-Based Multi-Platform Architecture: Azure backend.

6.2 Site Map (High Level)



7. Functional (FR) and Non-Functional (NFR) Requirements

7.1 Functional Requirements (Detailed)

The system comprises core functional requirements, expanded with descriptions, rationale, dependencies, and acceptance scenarios.

FR ID	Requirement Title	Priority
FR01	User Registration & Profile Management	Critical
FR02	Authentication & Secure Login	Critical
FR03	Symptom Entry & Risk Score Calculator	High
FR04	Intelligent Triage Recommendations	High
FR05	Online Appointment Booking System	Critical
FR06	Video Consultation Platform	High
FR07	Digital Prescription Generation	Medium

FR ID	Requirement Title	Priority
FR08	Automated Reminders	Medium
FR09	Emergency Access Override	High
FR10	Admin Dashboard & Reporting	Medium

FR01: User Registration & Profile Management

Description: The system shall provide a comprehensive registration interface allowing patients and dental practitioners to create accounts with role-based access. Users must provide essential identification details, contact information, and relevant medical history. The registration process includes email verification and password strength validation to ensure account security. Patient registration via phone/email with comprehensive profiles; Dentist registration with license verification and specialization details.

Rationale: Accurate user profiles are fundamental to delivering personalised healthcare services. Patient medical history informs risk assessment, whilst dentist credentials ensure qualified practitioners provide care. Secure registration prevents unauthorised access and maintains data integrity across the platform.

Dependencies: Email verification service integration, Database schema for user entities, Authentication framework (FR02), GDPR-compliant data storage.

Acceptance Scenarios:

- Scenario 1: New patient completes registration form with valid details and receives verification email within 2 minutes.
- Scenario 2: Dentist registration includes professional licence validation and clinic affiliation.
- Scenario 3: System rejects registration attempts with duplicate email addresses.

FR02: Authentication & Secure Login

Description: The system shall implement multi-factor authentication supporting password-based login, biometric verification, and optional two-factor authentication via SMS or authenticator apps. Session management includes automatic timeout after 30 minutes of inactivity and secure token refresh mechanisms. Password recovery follows industry-standard security protocols with time-limited reset links. Secure authentication with password reset capabilities; Role-based access control (patient, dentist, administrator).

Rationale: Healthcare data requires the highest level of security protection. Robust authentication prevents unauthorised access to sensitive patient records and ensures only verified users can book appointments or access consultation features. Multi-factor authentication significantly reduces the risk of account compromise.

Dependencies: User registration system (FR01), Encryption libraries for secure storage, SMS gateway for two-factor codes, OAuth integration for social login options.

Acceptance Scenarios:

- Scenario 1: User logs in with valid credentials and receives access within 3 seconds.
- Scenario 2: Failed login attempts (5+) trigger account lockout and security notification.
- Scenario 3: Password reset link expires after 24 hours and cannot be reused.

FR03: Symptom Entry & Risk Score Calculator

Description: The system shall provide an intuitive symptom checker interface where patients describe their dental concerns through guided questions, image uploads, and severity ratings. An algorithmic risk calculator processes inputs to generate a numerical risk score (0-10) categorising urgency as Low, Moderate, or High. The calculator considers pain intensity, duration, accompanying symptoms, and patient medical history. Guided questionnaire capturing pain level, swelling, bleeding, sensitivity, and duration; Interactive 3D tooth map for precise location identification.

Rationale: Accurate symptom assessment enables intelligent triage, ensuring patients with urgent conditions receive priority care whilst optimising dentist schedules. The risk score provides objective data for appointment prioritisation and helps patients understand the seriousness of their condition, reducing unnecessary emergency visits.

Dependencies: User profile with medical history (FR01), Clinical decision support algorithms, Image processing for symptom photos, Database of dental conditions and symptoms.

Acceptance Scenarios:

- Scenario 1: Patient reporting severe tooth pain with swelling receives High risk score (8-10) and immediate booking options.
- Scenario 2: Routine check-up request generates Low risk score (1-3) with standard appointment slots.
- Scenario 3: Risk calculator updates dynamically as patient answers additional questions.

FR04: Intelligent Triage Recommendations

Description: The system shall generate intelligent, personalized triage recommendations based on the calculated oral risk score. Depending on the severity level, the system will advise patients to monitor symptoms at home, schedule a routine dental appointment, or seek immediate urgent or emergency care.

Rationale: This requirement ensures that patients follow appropriate care pathways, prevents unnecessary clinic visits, and enables efficient utilization of dental resources by prioritizing high-risk cases.

Dependencies: FR03 – Symptom Entry & Risk Score Calculator.

Acceptance Scenarios:

- Scenario 1: High-risk cases are automatically routed to emergency or priority consultation workflows.
- Scenario 2: Low-risk cases receive self-care guidance and routine follow-up recommendations.

FR05: Online Appointment Booking System

Description: The system shall enable patients to view available appointment slots across multiple dentists and clinics, filtered by date, location, specialisation, and urgency level. The booking interface displays real-time availability, estimated wait times, and dentist profiles. Patients receive instant confirmation with calendar integration and the ability to modify or cancel appointments with appropriate notice periods. Search and filter dentists by specialty, location, language, and ratings; View real-time slots; Reschedule/cancel with notifications.

Rationale: Streamlined appointment booking reduces administrative overhead, eliminates phone tag, and empowers patients to choose convenient time slots. Real-time availability prevents double-booking conflicts. Integration with the risk calculator ensures urgent cases receive prioritised scheduling, improving patient outcomes and satisfaction.

Dependencies: Dentist availability management system, Risk score calculation (FR03), Scheduling algorithm engine, Calendar integration APIs, Notification system (FR08).

Acceptance Scenarios:

- Scenario 1: Patient with High risk score sees priority slots within next 24 hours.
- Scenario 2: System prevents booking conflicts by locking slots during selection process.
- Scenario 3: Appointment confirmation email includes clinic address, dentist details, and preparation instructions.

FR06: Video Consultation Platform

Description: The system shall provide an integrated video consultation platform supporting high-definition video, secure audio, screen sharing, and real-time chat. Dentists can visually assess conditions, provide preliminary advice, and determine if in-person visits are necessary. The platform includes recording capabilities (with consent) for documentation purposes and supports up to 4 participants for multi-disciplinary consultations. Synchronous video calls and asynchronous store-and-forward with photo/X-ray uploads; In-session chat and file exchange.

Rationale: Remote consultations expand access to dental care, particularly for patients in rural areas or with mobility limitations. Virtual triage reduces unnecessary clinic visits, optimises dentist time, and provides patients with professional guidance without travel costs. Video documentation creates valuable records for treatment planning and follow-up care.

Dependencies: Appointment booking system (FR05), Secure authentication (FR02), High-bandwidth infrastructure, HIPAA-compliant video service, Patient consent management.

Acceptance Scenarios:

- Scenario 1: Patient joins consultation 5 minutes early and enters virtual waiting room.
- Scenario 2: Video quality automatically adjusts based on available bandwidth without disconnection.
- Scenario 3: Consultation recording is encrypted and stored in patient record with audit trail.

FR07: Digital Prescription Generation

Description: The system shall enable dentists to create, sign, and transmit digital prescriptions directly from the consultation interface. Prescriptions include medication details, dosage instructions, duration, and potential side effects. The platform checks for drug interactions against patient medical history and provides electronic transmission to selected pharmacies. Patients can view prescription history and set medication reminders. Electronic prescriptions with diagnosis notes; PDF generation; Integration with reminders.

Rationale: Digital prescriptions eliminate transcription errors from illegible handwriting, prevent medication conflicts through automated checks, and streamline pharmacy workflows. Electronic transmission reduces patient wait times and ensures accurate medication dispensing. Prescription history provides valuable data for continuity of care and treatment effectiveness analysis.

Dependencies: Video consultation platform (FR06), User medical history (FR01), Drug interaction database, Electronic signature validation, Pharmacy integration APIs.

Acceptance Scenarios:

- Scenario 1: Dentist prescribes antibiotics and system alerts about patient penicillin allergy.
- Scenario 2: Prescription transmits to pharmacy within 5 minutes of dentist signature.
- Scenario 3: Patient receives mobile notification with medication instructions and pickup details.

FR08: Automated Reminders

Description: The system shall automatically generate medication and follow-up reminders by intelligently parsing digital prescriptions. Notifications shall be delivered through multiple channels, including push notifications and SMS, with escalation mechanisms for missed doses or appointments.

Rationale: Automated reminders improve patient adherence to prescribed treatments and follow-up schedules, leading to better clinical outcomes and reduced treatment failures.

Dependencies: FR07 – Digital Prescription Generation.

Acceptance Scenarios:

- Patients receive daily medication reminders at scheduled times.
- Missed doses trigger follow-up alerts or escalation notifications.

FR09: Emergency Access Override

Description: The system shall provide a 24/7 emergency access feature enabling patients to immediately connect with available dental professionals through hotline or teleconsultation. The system shall transmit essential patient context, including identification, medical history, and risk score, to ensure rapid and informed response.

Rationale: This feature ensures timely intervention during urgent dental situations, minimizing complications and improving patient safety.

Dependencies: FR05 – Online Appointment Booking System, FR03 – Symptom Entry & Risk Score Calculator.

Acceptance Scenarios:

- Activating the emergency button connects the patient to a dentist within one minute.
- Dentists receive patient context before initiating the consultation.

FR10: Admin Dashboard & Reporting

Description: The system shall provide an administrative dashboard for managing users, clinics, system settings, audit logs, and generating analytical reports related to system usage and performance.

Rationale: Administrative oversight ensures system reliability, security monitoring, and data-driven decision-making for continuous improvement.

Dependencies: All functional requirements (FR01–FR09).

Acceptance Scenarios:

- Administrators can generate system usage and performance reports in under 10 seconds.
- Audit logs accurately record all critical system activities.

7.2 Non-Functional Requirements (Detailed)

NFR ID	Requirement Title	Description
NFR01	System Security & Data Protection	Implement end-to-end encryption (AES-256), RBAC, audits; Comply with GDPR, HIPAA.
NFR02	Performance & Response Time	≤2s load time, ≥10,000 concurrent users; Sub-100ms DB queries.
NFR03	Reliability & Uptime	>99.9% uptime, auto-recovery.

NFR ID	Requirement Title	Description
NFR04	Usability & Accessibility	WCAG 2.1 compliance, intuitive UI.
NFR05	Scalability	Horizontal scaling, cloud-ready (Azure).
NFR06	Compatibility	Android/web, low-bandwidth support.

NFR01: System Security & Data Protection

Description: Comprehensive security measures including end-to-end encryption for all data transmission, AES-256 encryption for data at rest, role-based access control (RBAC), and regular security audits. All sensitive data must comply with GDPR, HIPAA, and relevant healthcare data protection regulations. The system shall maintain detailed audit logs of all data access and modifications, with automatic alerts for suspicious activities.

Rationale: Healthcare data represents one of the most sensitive information categories, requiring the highest security standards. Data breaches can result in severe financial penalties, legal consequences, and irreparable damage to patient trust. Robust security measures protect patient privacy, ensure regulatory compliance, and maintain the integrity of clinical records essential for quality care delivery.

Acceptance Scenarios:

- Scenario 1: Security audit reveals zero unauthorised access attempts succeeded in past quarter.
- Scenario 2: Encrypted patient data remains unreadable even if database is compromised.
- Scenario 3: System blocks dentist from accessing patient records outside their assigned cases.
- Scenario 4: Automated penetration testing identifies and reports vulnerabilities within 24 hours.
- Scenario 5: All API endpoints require valid authentication tokens that expire after 1 hour.

NFR02: Performance & Response Time

Description: Deliver optimal performance with page loads under 2 seconds, search results within 1 second, appointment bookings in under 3 seconds, video consultations initiating within 5 seconds. Support 10,000 concurrent users with horizontal scaling. Database queries optimised with sub-100ms response times.

Rationale: Performance directly impacts user satisfaction and system adoption. Slow response times frustrate users, particularly patients in pain seeking urgent care. Fast, responsive interfaces enhance trust and efficiency, whilst scalable architecture ensures consistent performance as the user base grows. Performance optimisation reduces infrastructure costs and improves overall system reliability.

Acceptance Scenarios:

- Scenario 1: Dashboard loads completely within 2 seconds on standard broadband connection.
- Scenario 2: System maintains sub-3-second response times during peak usage hours (8am-6pm).
- Scenario 3: Video consultation maintains stable connection with less than 100ms latency.
- Scenario 4: Search for available appointments returns results in under 800ms.
- Scenario 5: System automatically scales to handle 50% traffic spike without manual intervention.

NFR03: Reliability & Uptime

Description: Achieve >99.9% uptime with automatic failover and recovery mechanisms. Implement redundant systems and regular backups.

Rationale: Ensures continuous availability for healthcare services, critical in emergencies.

Acceptance Scenarios:

- Scenario 1: System recovers from server failure within 5 minutes.
- Scenario 2: Daily backups tested for restorability.
- Scenario 3: Uptime monitored and reported monthly.

NFR04: Usability & Accessibility

Description: Comply with WCAG 2.1 standards, providing intuitive interfaces, mobile responsiveness, and support for low digital literacy users.

Rationale: Enhances user adoption, especially in rural areas.

Acceptance Scenarios:

- Scenario 1: Screen reader compatibility tested.
- Scenario 2: User satisfaction score >4/5 in usability tests.
- Scenario 3: Interfaces support Bengali language.

NFR05: Scalability

Description: Support horizontal scaling on cloud infrastructure (e.g., Azure), handling growth to millions of users.

Rationale: Prepares for national expansion.

Acceptance Scenarios:

- Scenario 1: System handles 2x load without performance degradation.
- Scenario 2: Auto-scaling rules trigger appropriately.
- Scenario 3: Cost-effective scaling demonstrated.

NFR06: Compatibility

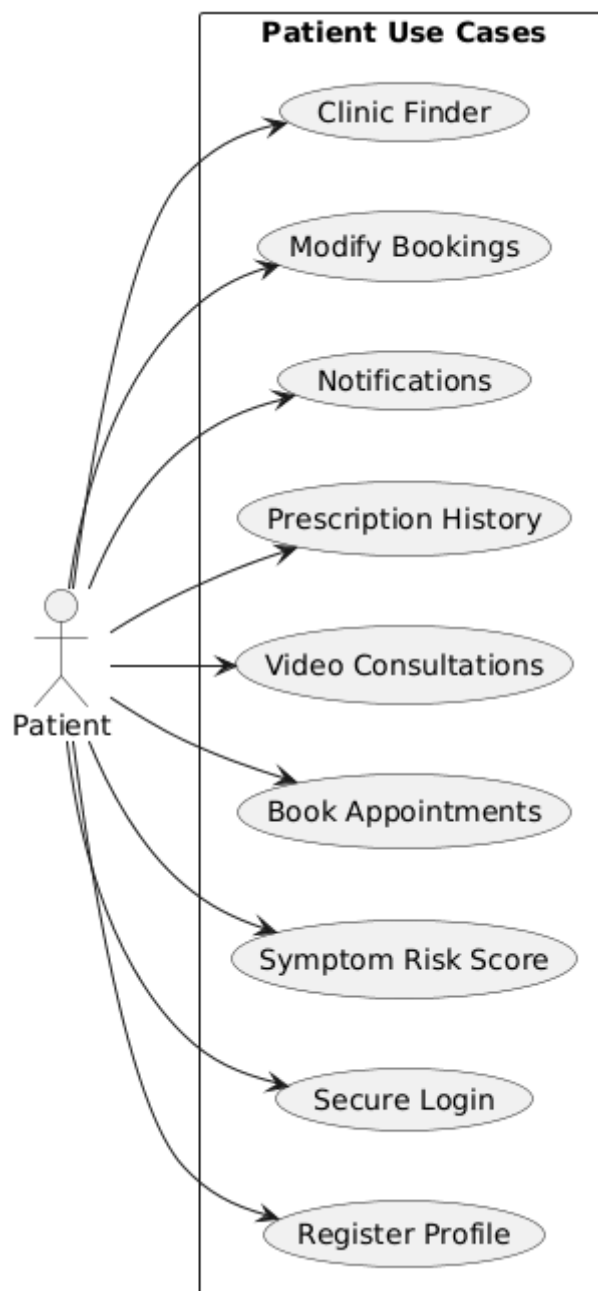
Description: Compatible with Android/web browsers, optimized for low-bandwidth connections.

Rationale: Addresses connectivity issues in Bangladesh.

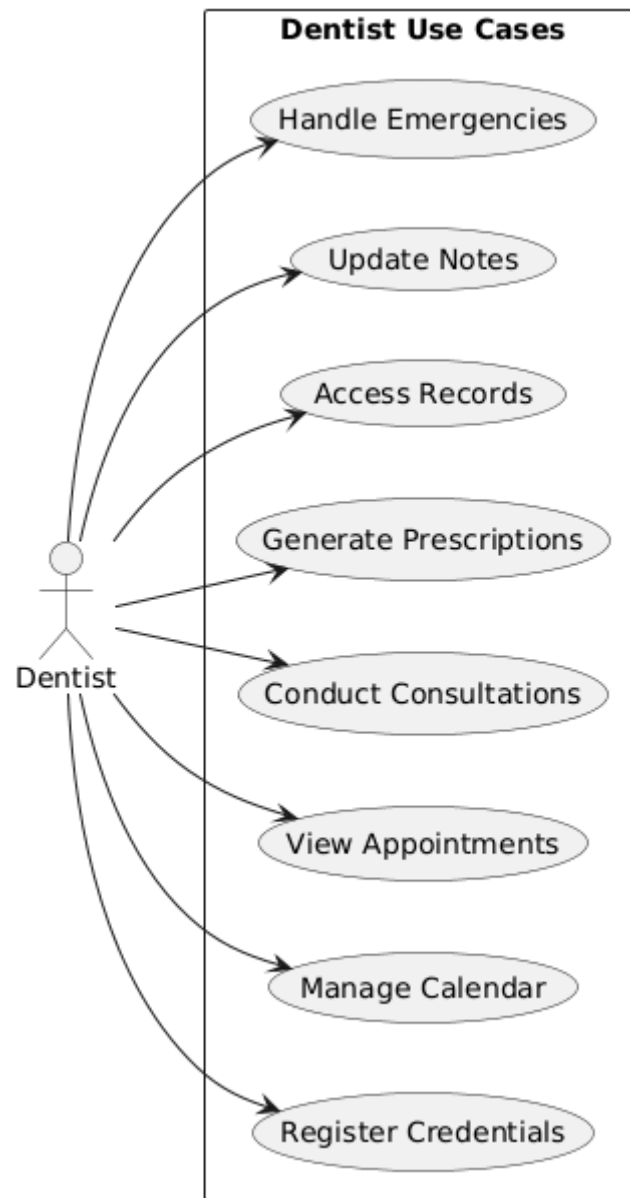
Acceptance Scenarios:

- Scenario 1: Functions on Android 8+ and major browsers.
- Scenario 2: Offline mode for clinic finder works without internet.
- Scenario 3: Graceful degradation on slow networks.

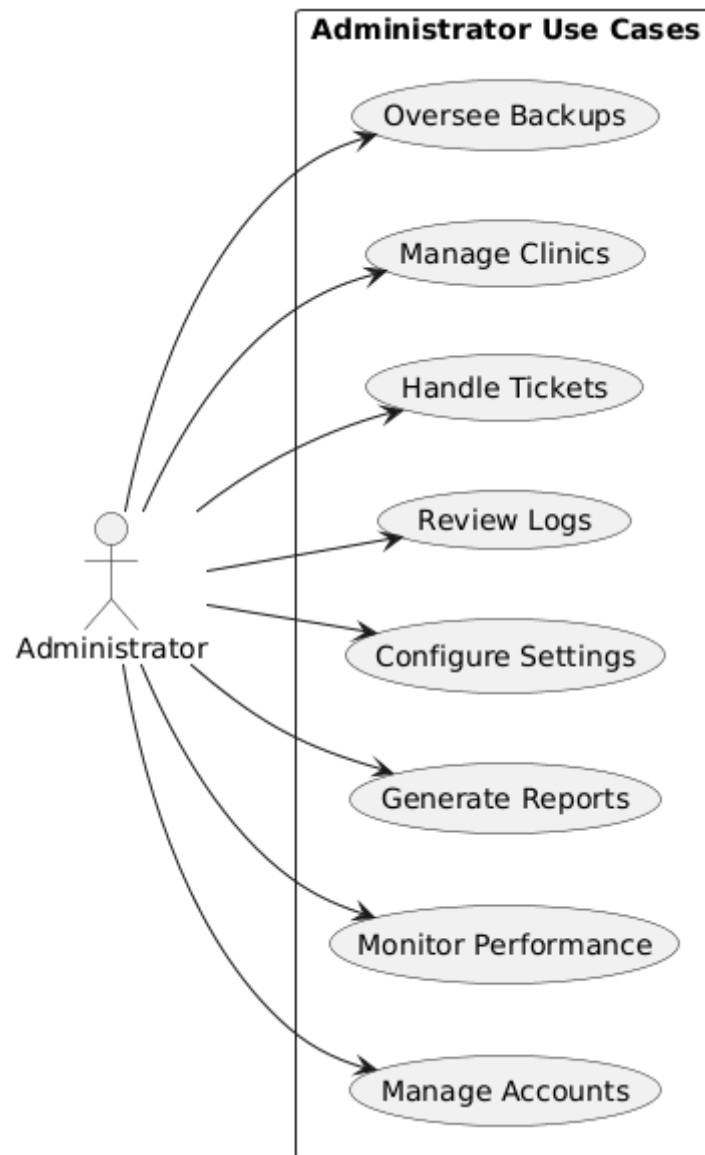
8. Use Case Diagram



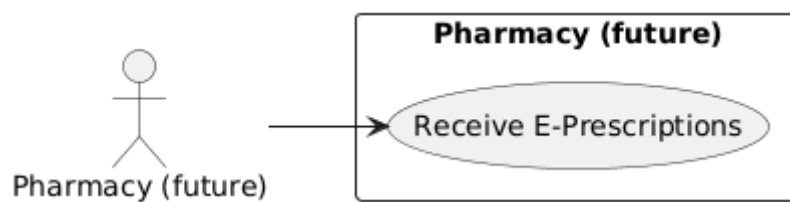
Use case diagram 1: Patient



Use case diagram 2: Dentist



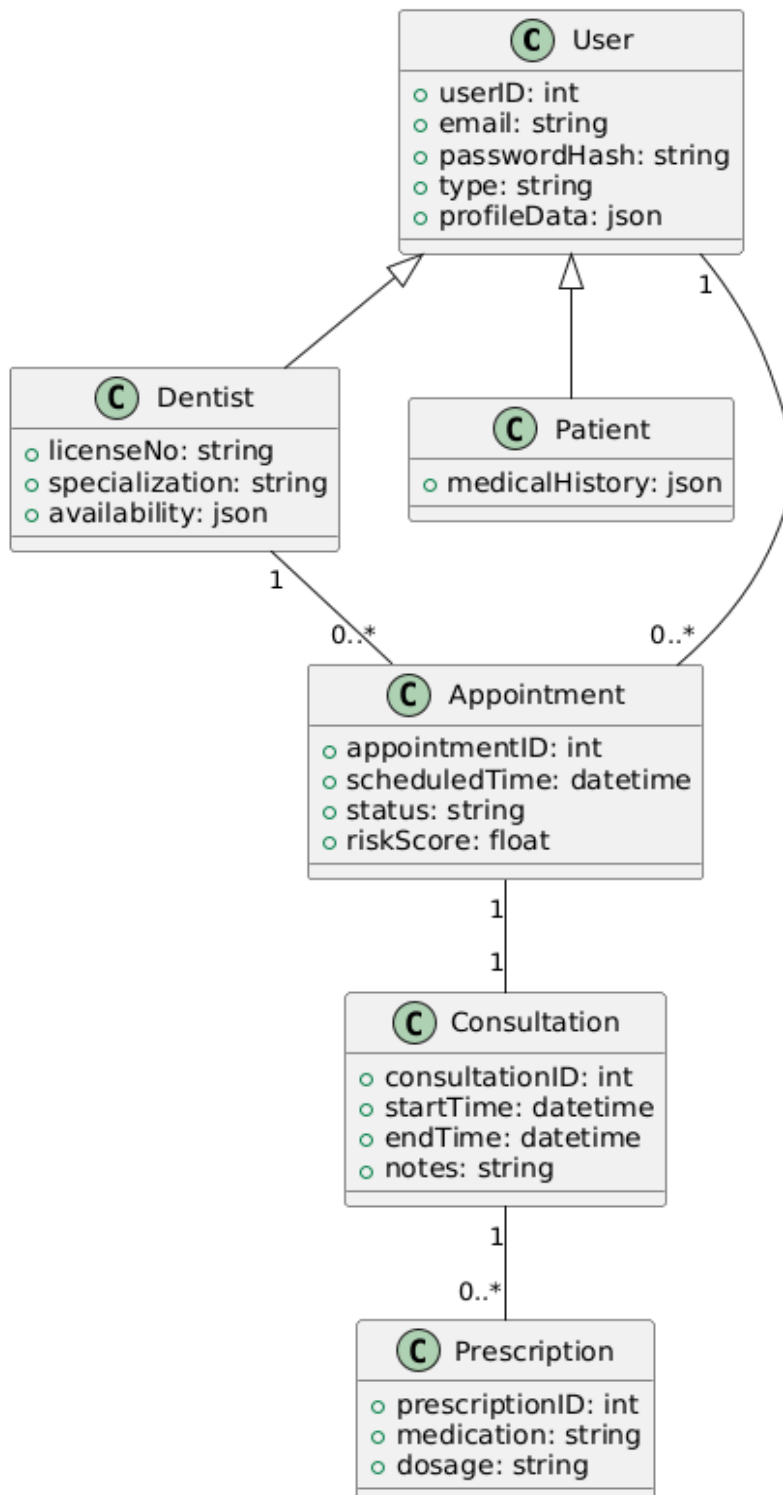
Use case diagram 3: Administrator



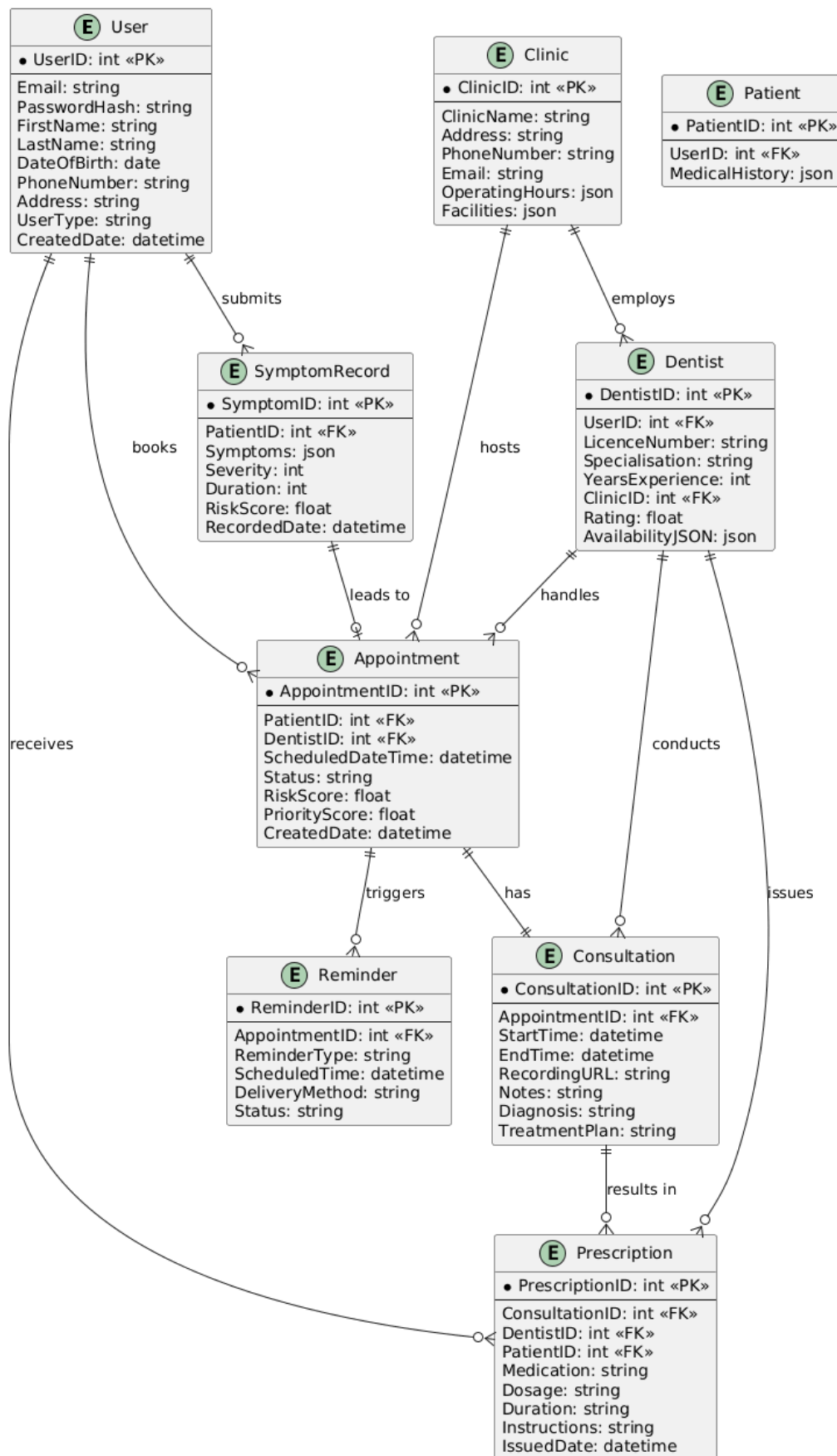
Use case diagram 4: Pharmacy (future)

9. Class Diagram (+ ER Diagram)

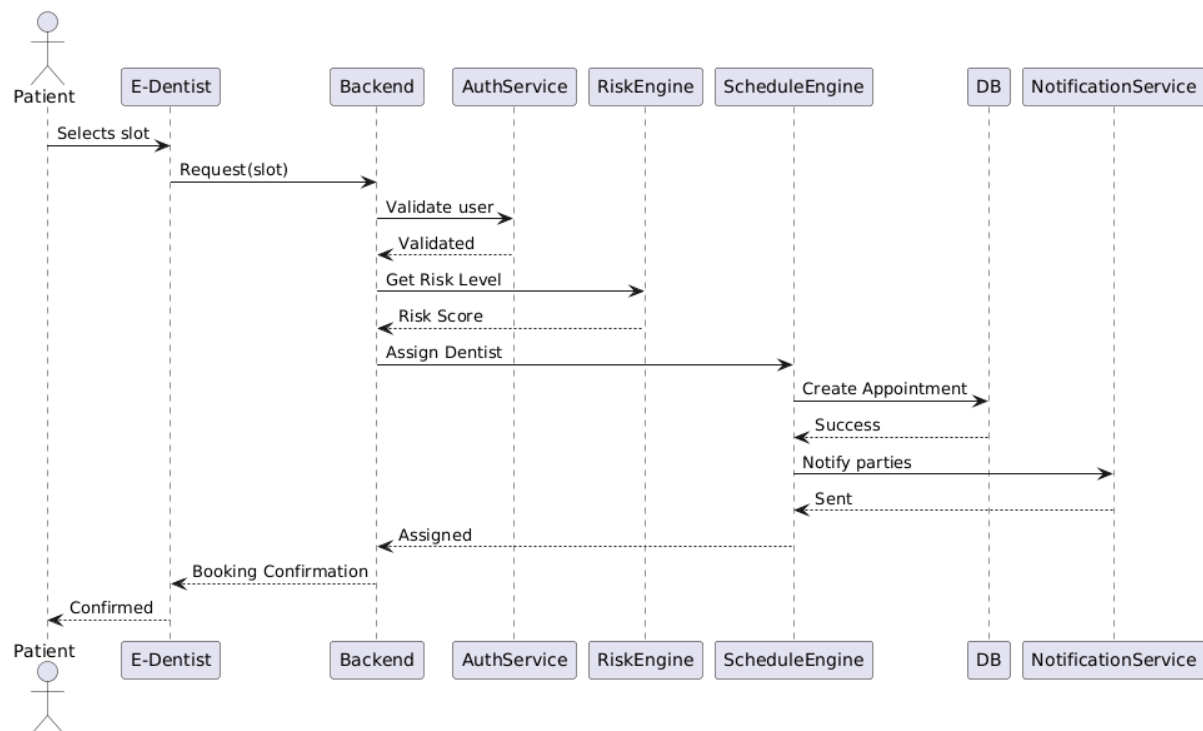
9.1 Class Diagram



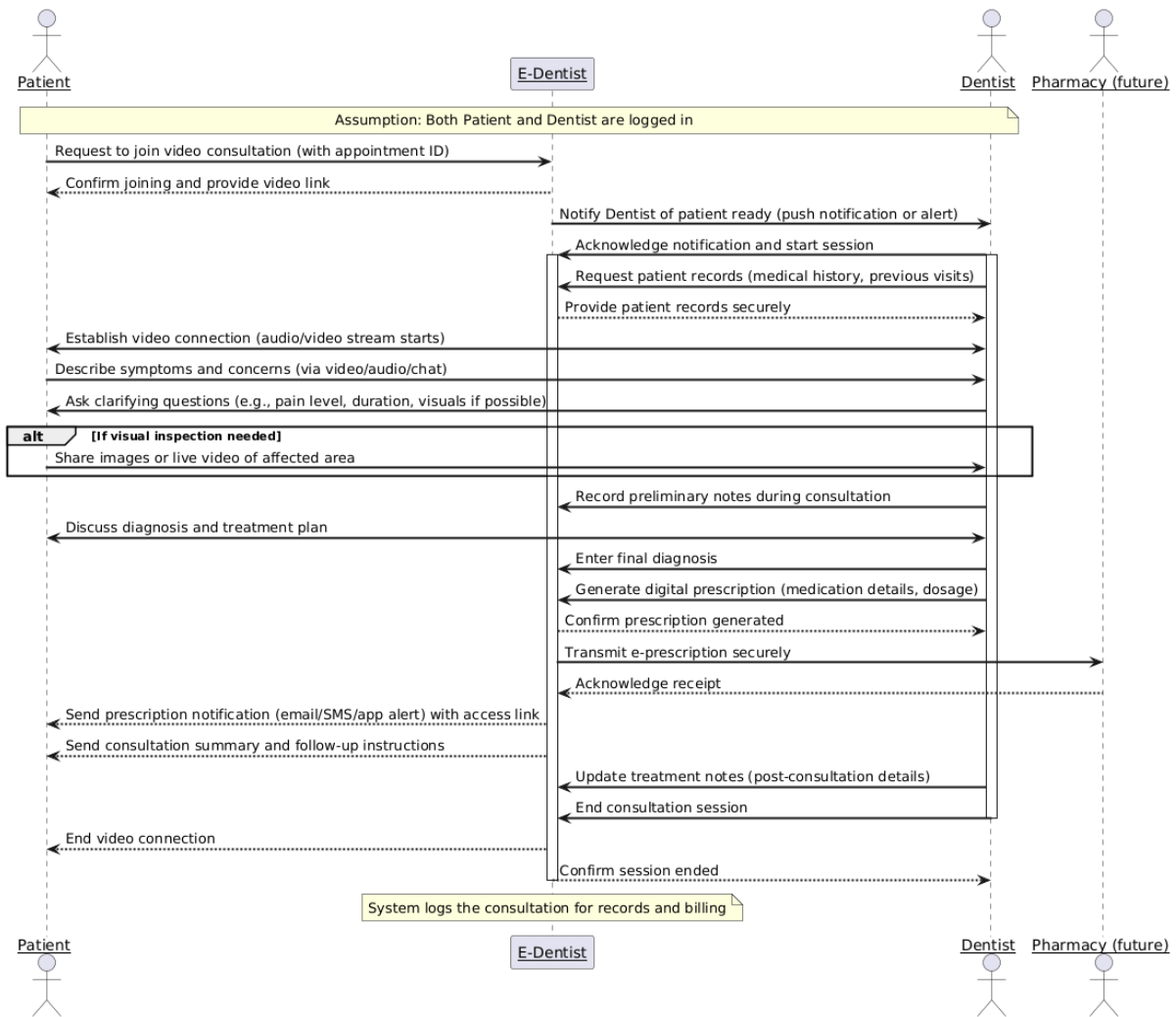
9.2 ER Diagram



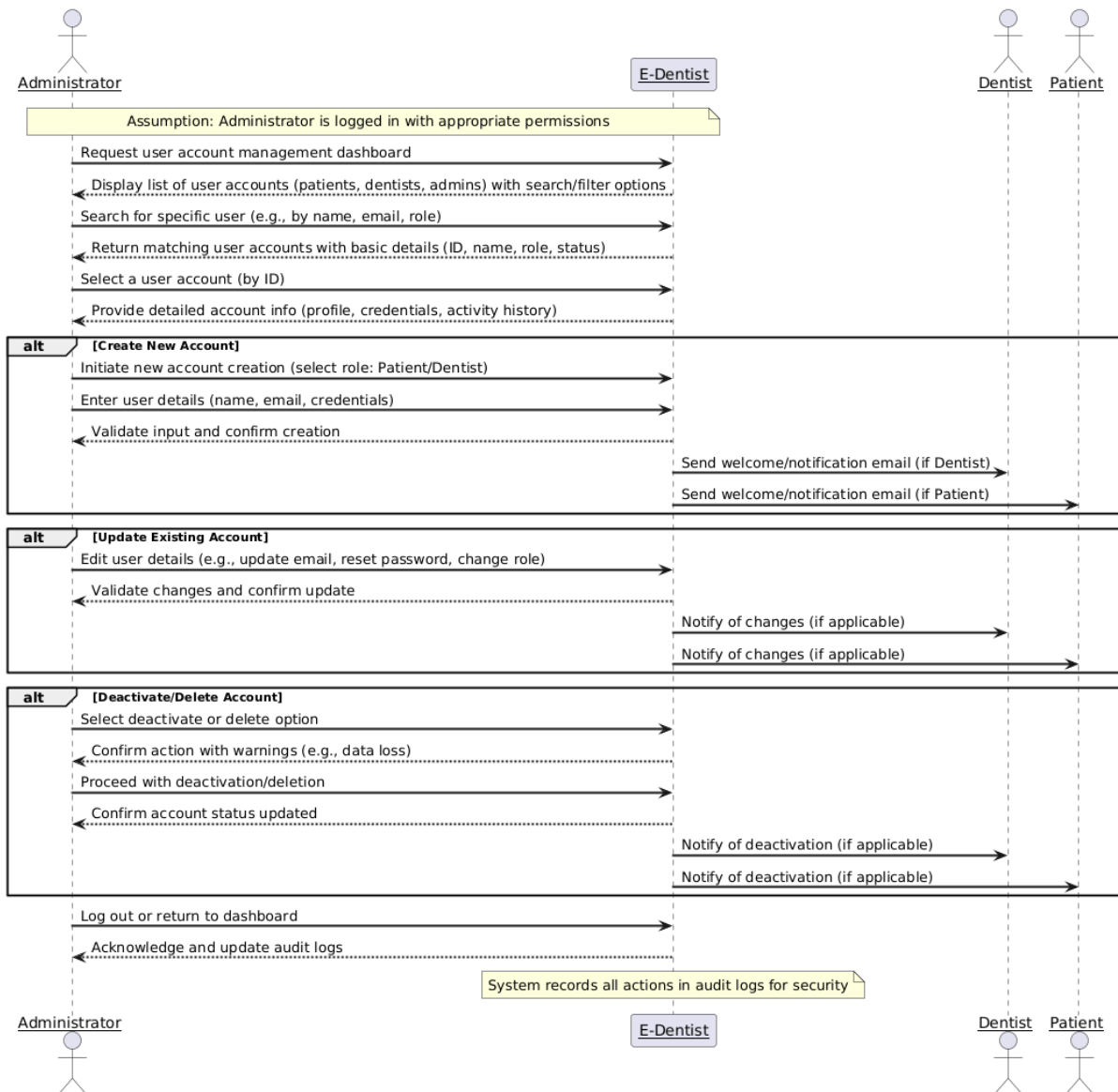
10. Sequence Diagram



Sequence Diagram 1: Appointment Booking

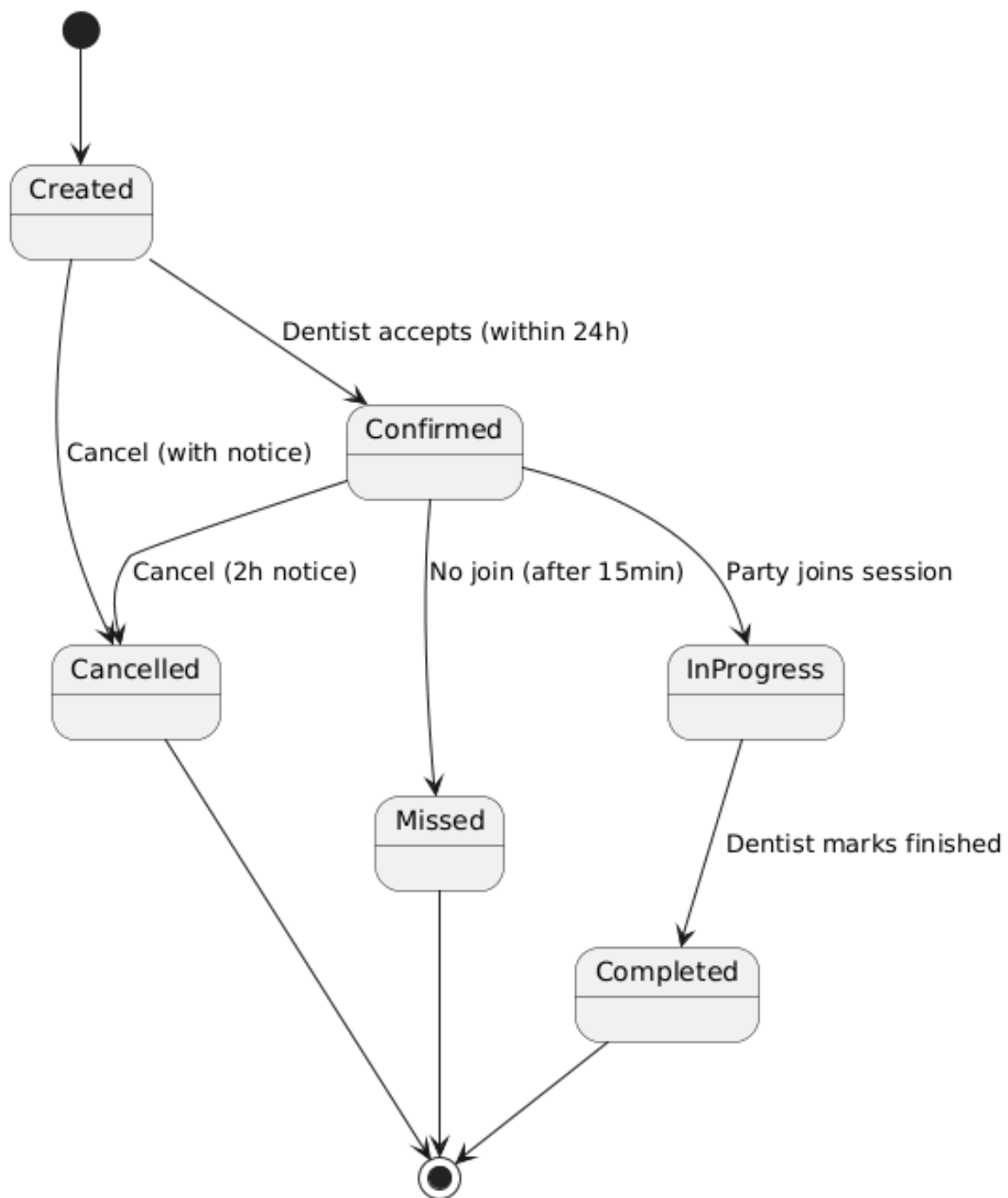


Sequence Diagram 2: Consultation Process

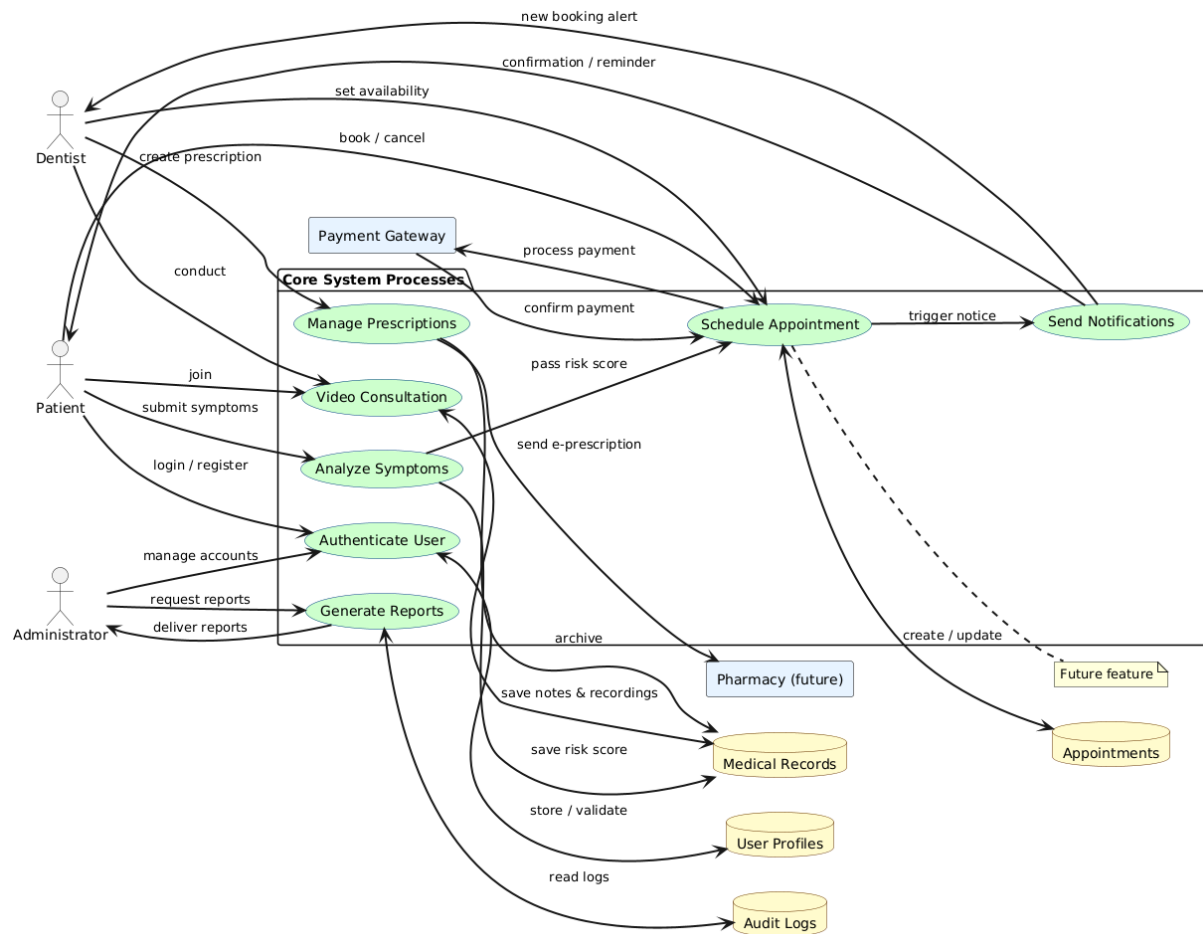


Sequence Diagram 3: Administrator Manage Accounts

11. State Diagram – Appointment Lifecycle



12. Data Flow Diagram (Level 1)



13. Priority-Based Dynamic Scheduling Algorithm

13.1 Algorithm Overview

The scheduling algorithm assigns appointments by balancing risk scores, availability, preferences, emergency flags, and load balancing.

Steps:

1. Risk Assessment: Determine priority from symptom checker.
2. Availability Check: Query open slots.
3. Priority Calculation: Composite score from risk, flags, etc.
4. Load Balancing: Adjust for dentist workload.
5. Conflict Resolution: Resolve using scores.
6. Confirmation & Notification: Notify parties.

13.2 Rules & Priority Scoring

Risk Level	Priority Points	Scheduling Notes
High (8-10)	+5	Same-day/next-day; emergency override
Moderate (4-7)	+3	Within 3-7 days; consider preferences
Low (1-3)	+1	Within 2 weeks; routine
First-time patient	+1	Extra time for assessment
Follow-up visit	+2	Prefer original dentist
Emergency flag	+7	Immediate; overrides queue

Workload Adjustment: +1 (0-60% load), 0 (61-85%), -2 (86%+).

13.3 Pseudo-Code

```

FUNCTION ScheduleAppointment(patient, requestedDateTime, symptoms):
    riskScore = CalculateRiskScore(symptoms, patient.medicalHistory)
    priorityPoints = 0
    IF riskScore >= 8 THEN priorityPoints += 5
    ELSE IF riskScore >= 4 THEN priorityPoints += 3
    ELSE priorityPoints += 1

    IF patient.isEmergency THEN priorityPoints += 7
    IF patient.isFirstTime THEN priorityPoints += 1
    IF patient.isFollowUp THEN priorityPoints += 2

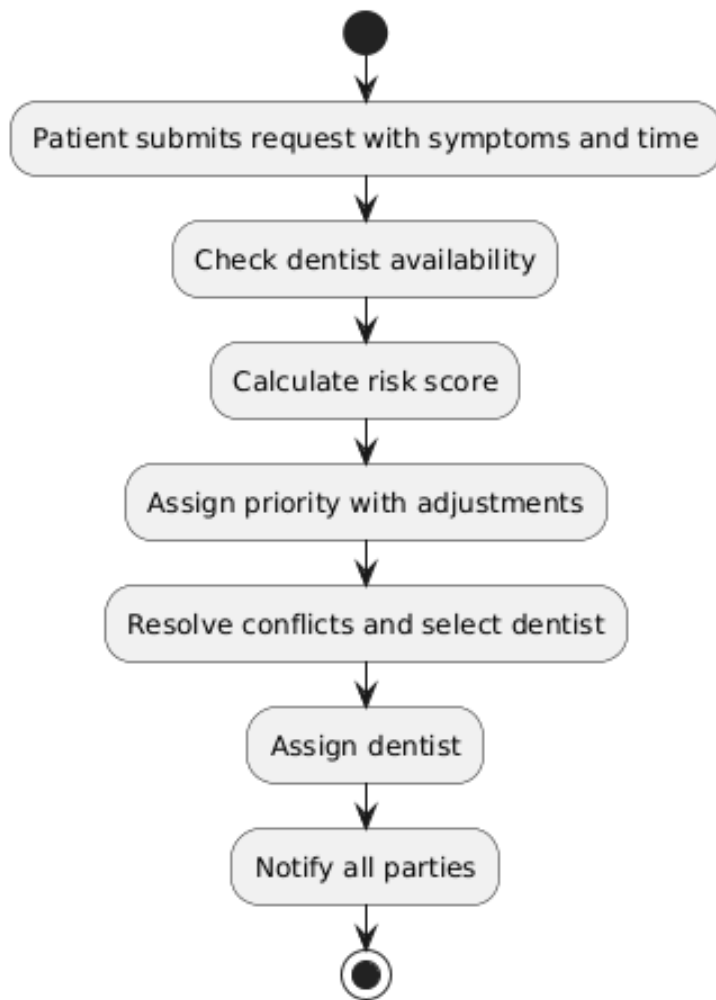
    availableDentists = QueryDentistAvailability(requestedDateTime)
    IF availableDentists.isEmpty() THEN
        suggestedSlots = FindNextAvailableSlots(requestedDateTime)
        RETURN suggestedSlots
    END IF

```

```
FOR EACH dentist IN availableDentists:
    workloadFactor = CalculateWorkloadFactor(dentist)
    dentist.adjustedPriority = priorityPoints + workloadFactor
    IF DentistSpecialisationMatches(dentist, symptoms) THEN
        dentist.adjustedPriority += 2
    END IF
END FOR

selectedDentist = dentist WITH MAX(adjustedPriority)
appointment=CreateAppointment(patient, selectedDentist, requestedDateTime)
SendNotifications(patient, selectedDentist, appointment)
RETURN appointment
END FUNCTION
```

13.4 Workflow (Activity Diagram)



14. Requirements Traceability Matrix

To ensure compliance with CMMI V3.0 Requirements Development & Management (RDM) and Process Quality Assurance (PQA), a traceability matrix is included. This matrix traces functional requirements (FRs) to use cases, dependencies, acceptance scenarios, and related diagrams.

FR ID	Related Use Cases	Dependencies	Acceptance Scenarios	Related Diagrams
FR01	Register Account	None	1-3	Class Diagram (User), ER Diagram (User)
FR02	Authenticate	FR01	1-3	Sequence Diagram (Auth Service)

FR ID	Related Use Cases	Dependencies	Acceptance Scenarios	Related Diagrams
FR03	Enter Symptoms, Calculate Risk Score	FR01	1-3	Data Flow Diagram (Symptom Analysis), Algorithm Pseudo-Code
FR04	Intelligent Triage Recommendations	FR03	High/Low risk scenarios	State Diagram (Appointment Lifecycle)
FR05	Book Appointment, Manage Availability	FR03, FR02	1-3	Sequence Diagram (Appointment Booking), DFD (Appointment Scheduling)
FR06	Conduct Consultation	FR05, FR02	1-3	Sequence Diagram, Use Case Diagram (Join/Conduct Consultation)
FR07	Generate Prescription	FR06, FR01	1-3	Class Diagram (Prescription), ER Diagram (Prescription)
FR08	Automated Reminders	FR07	Daily/escalation	DFD (Notification Dispatch), ER Diagram (Reminder)
FR09	Emergency Access Override	FR05, FR03	Connect <1 min	Use Case Diagram (Handle Emergency Requests)
FR10	Manage Users, Generate Reports	All FRs	Reports <10s	Use Case Diagram (Admin tasks), DFD (Reporting & Analytics)

This matrix ensures bidirectional traceability, supporting verification, validation, and change management.

15. Key Benefits

- Faster Patient Access: Reduced waiting times.
- Intelligent Triage: Prioritizes urgent cases.
- Emergency Readiness: Rapid response.
- Better Dental Outcomes: Personalized plans.
- Efficient Remote Workflow: Less admin burden.

16. System Architecture & Key Actors

Primary Actors: Patient, Dental Professional, System Administrator. External Integrations: Payment gateways (future), SMS/push notifications, Maps, Azure cloud.

17. Compliance with CMMI on Analysis & Documentation

E-Dentist adheres to CMMI-DEV V3.0 Maturity Level-3 ("Defined") standards, ensuring process maturity and documentation quality. At Level 3, processes are tailored from organizational standards, with complete practice group implementations in applicable Practice Areas (PAs). This SRS demonstrates compliance through structured documentation, diagrams, and traceability.

Compliance Mapping (Focused on DEV View PAs; all Level 3 practices addressed where applicable):

- **Requirements Development & Management (RDM):** Stakeholder needs elicited via interviews and literature review (Section 4); FR/NFR detailed with rationale, dependencies, scenarios (Section 7); Traceability matrix (Section 14); Validation via acceptance scenarios.
- **Technical Solution (TS):** Designs/solutions in UML diagrams (Sections 8-12), class/ER diagrams (Section 9), algorithm (Section 13); Alternatives considered (e.g., cloud vs. on-prem in vision).
- **Product Integration (PI):** Defined integration points (video API, pharmacy API, Azure backend); Sequence diagram shows integration flow (Section 10).
- **Verification and Validation (VV):** Acceptance scenarios/test criteria (Section 7); Peer review implied in document iterations; Ensures patient safety/clinical outcomes via risk-based features.
- **Configuration Management (CM):** SRS version control (this is v3.0 final); Baselines for requirements; Change control via dependencies/traceability.
- **Process Quality Assurance (PQA):** Audit logs in FR10; Compliance mapping ensures objective evaluation; Improvement opportunities noted (e.g., future interoperability).
- **Risk and Opportunity Management (RSK):** Constraints/risks identified (Section 5.4); Mitigation via offline features, security NFRs; Opportunities in vision (e.g., insurance integration).
- **Decision Analysis and Resolution (DAR):** Structured decisions in algorithm (Section 13); Criteria for FR priorities; Alternatives in designs (e.g., sync/async consultations).
- **Planning (PLAN):** Project scope in introduction/motivation; Dependencies in FRs; Estimates implied in NFR02 (performance).
- **Monitor and Control (MC):** Reporting in FR10; Corrective actions via reminders/emergency overrides.
- **Estimating (EST):** Scalability NFR05; Effort estimates in acceptance scenarios (e.g., response times).
- **Governance (GOV):** Senior management sponsorship implied in vision; Resources for development (e.g., Azure).
- **Implementation Infrastructure (II):** Organizational processes via standards (WCAG, HIPAA); Training in OT mapping.

- **Organizational Training (OT):** User training for low-literacy (constraints); Strategic plans in vision.
- **Process Asset Development (PAD):** Reusable assets like UML templates; Library of diagrams.
- **Process Management (PCM):** Improvements via gap analysis (Section 5.2); Deployment of standards.
- **Causal Analysis and Resolution (CAR):** Root cause for missed appointments (state diagram); Improvements submitted via admin reports.
- **Peer Reviews (PR):** Reviews of requirements/diagrams during development.
- **Data Management (DM) & Data Quality (DQ):** Medical history/patient data handling (FR01, NFR01); Quality via risk calculator validation.
- **Workforce Empowerment (WE):** Role-based access empowers users; Competencies for dentists (license verification).
- **Enabling Security (ESEC) & Managing Security Threats & Vulnerabilities (MST):** NFR01 covers encryption, audits, threat mitigation.

This comprehensive mapping ensures the software lifecycle processes meet international quality and engineering standards for CMMI Level 3, with tailored organizational processes, quantitative management where applicable, and continuous improvement.

18. UX/UI Design Guidelines and Standards

This section defines the UX and UI standards, interaction principles, and usability metrics used to design the E-Dentist experience. The guidelines are applied consistently across all screens (mobile and web) to ensure safety, clarity, efficiency, and a predictable user experience.

Guideline 1: The UX Standard (Behavioral Logic)

STANDARD TO FOLLOW: NIELSEN NORMAN GROUP
(NN/G) — JAKOB NIELSEN'S 10 USABILITY HEURISTICS.

E-dentist application rules:

1. Visibility of system status: Always provide feedback for loading, saving, success, and failure states.
2. Match between system and real world: Use clinic-friendly terminology and workflows (Patient, Visit, Prescription, Invoice).
3. User control and freedom: Provide Back/Cancel/Close consistently; allow undo where feasible; confirm destructive actions.
4. Consistency and standards: Reuse the same layout patterns, component behaviors, and labels across the product.
5. Error prevention: Prefer constraints and inline validation; disable actions when inputs are incomplete or invalid.
6. Recognition rather than recall: Surface recent items, templates, and suggested defaults to reduce memory load.
7. Flexibility and efficiency: Support shortcuts for frequent tasks (search, recent patients, repeat actions).

8. Aesthetic and minimalist design: Keep screens focused on the primary task and remove non-essential UI noise.
9. Help users recognize, diagnose, and recover from errors: Provide plain-language messages and clear next actions (Retry, Edit, Contact Support).
10. Help and documentation: Provide micro-help in context (empty states, tooltips, quick tips) when users need guidance.

Guideline 2: The UI Standard (Visual System)

STANDARD TO FOLLOW: MATERIAL DESIGN 3 (M3) BY GOOGLE.

E-Dentist design system rules:

1. Define reusable foundations (design tokens): color, typography, spacing, radius, elevation, and motion.
2. Use M3 components as the baseline (buttons, cards, dialogs, text fields, chips, navigation) and keep custom components consistent with M3 patterns.
3. Design and document component states: default, pressed/active, disabled, loading, and error.
4. Maintain clear visual hierarchy: one primary action per screen/section and predictable placement of secondary actions.
5. Ensure touch/click targets are comfortably usable and consistent across devices.

Guideline 3: Interaction Science (Efficiency)

STANDARDS TO FOLLOW: FITTS'S LAW, HICK'S LAW, AND MILLER'S LAW.

E-Dentist application rules:

1. Fitts's Law: Place primary actions in easy-to-reach areas and size targets to reduce mis-taps and speed up interaction.
2. Hick's Law: Reduce decision time by grouping options, using progressive disclosure, and avoiding overloaded menus.
3. Miller's Law: Reduce working-memory burden by chunking information, breaking long forms into sections/steps, and using templates and defaults.

Guideline 4: Usability Standard (Definition and Measurement)

STANDARD TO FOLLOW: ISO 9241-11 — USABILITY (EFFECTIVENESS, EFFICIENCY, SATISFACTION) IN A SPECIFIED CONTEXT OF USE.

Usability evaluation metrics for E-Dentist:

1. Effectiveness: task success rate, completion accuracy, and error rate.
2. Efficiency: time-on-task, number of taps/clicks, steps to complete, and backtracking frequency.
3. Satisfaction: SUS (System Usability Scale) or short ease-of-use ratings, plus qualitative feedback themes.

Guideline 5: Interaction Principles (Dialogue Quality)

STANDARD TO FOLLOW: ISO 9241-110 — DIALOGUE PRINCIPLES FOR HUMAN-SYSTEM INTERACTION.

E-Dentist dialogue principles:

4. Suitability for the task: workflows are optimized for real clinical tasks with minimal unnecessary steps.
5. Self-descriptiveness: each screen clearly communicates where the user is and what to do next.
6. Conformity with user expectations: common platform patterns are respected to keep behavior predictable.
7. Learnability: consistent patterns are repeated so users learn once and reuse everywhere.
8. Controllability: the user controls pace, can exit flows, and can correct mistakes easily.
9. Error tolerance: errors are prevented where possible, detected early, and recoverable with clear actions.
10. Engagement: calm, readable visual design that supports focus and trust in clinical contexts.

Guideline 6: Accessibility Baseline

STANDARD TO FOLLOW: WCAG 2.2 (RECOMMENDED BASELINE FOR ACCESSIBLE INTERACTION).

E-Dentist accessibility rules:

11. Readable typography and sufficient contrast for text and key UI elements.
12. Clear labels for inputs and error messages directly associated with the relevant fields.
13. Logical focus order and visible focus indicators (web).
14. Adequate target sizes and spacing to support accurate tapping/clicking.

19. References

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Appendix A: Scope, Assumptions, and Clarifications

A.1 System Scope

The E-Dentist system provides a tele-dentistry platform enabling online consultations, intelligent rule-based triage, appointment scheduling, digital prescriptions, automated reminders, and clinic navigation services. The system supports Patients, Licensed Dentists, and Administrators through web and mobile interfaces.

A.2 Out-of-Scope

- Automated clinical diagnosis without dentist validation
- Medication delivery and logistics
- Insurance claim processing
- Fully autonomous AI-based decision making

A.3 Assumptions & Dependencies

- Users have access to smartphones or web-enabled devices
- Dentists are externally licensed and verified
- Internet connectivity may be intermittent
- Risk scoring logic is rule-based and clinically approved
- SMS, Video, and Map services rely on third-party APIs

A.4 Clarification on Intelligent Components

All intelligent features in E-Dentist are implemented as rule-based clinical decision support systems.

Machine learning-based analytics are considered future enhancements and are not part of the current scope.

A.5 Verification Methods Summary

- Functional Requirements: Acceptance Scenarios & Use Case validation
- Performance NFRs: Load testing (e.g., JMeter)
- Security NFRs: OWASP-based penetration testing
- Usability: WCAG 2.1 compliance testing

A.6 Algorithm Correctness Note

The priority-based scheduling algorithm operates in $O(n)$ time complexity, where n represents the number of available dentists. Deterministic priority scoring ensures predictable scheduling outcomes.